

Manual

Failure Mode Analysis / Measure Tracking REK-FSM 8.1

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Contents

1	Overview of Failure Mode Analysis/Measure Tracking	4
2	Failure Analysis of Complaints	5
3	Measure Tracking	10

1 Overview of Failure Mode Analysis/Measure Tracking

Fields of application

Besides the graphic analysis of the failures leading to a complaint (failure type, location, cause and causer), this function also allows for measures assigned to a complaint to be modified centrally.

Implementation notes

This component is recommendable, provided that the measures recorded for complaints are to be tracked and edited centrally. It is possible, for example, to filter all measures by specific responsible persons taking into account the status of measures.

Another field of application applies if the failures assigned to a complaint are required to be analyzed in detail (in graphics).

Measures and failures need to be collected in complaint management to be able to use this function.

Integration

This component is basically connected with the component for collection, modification and management of complaints using failure and measure recording.

Features

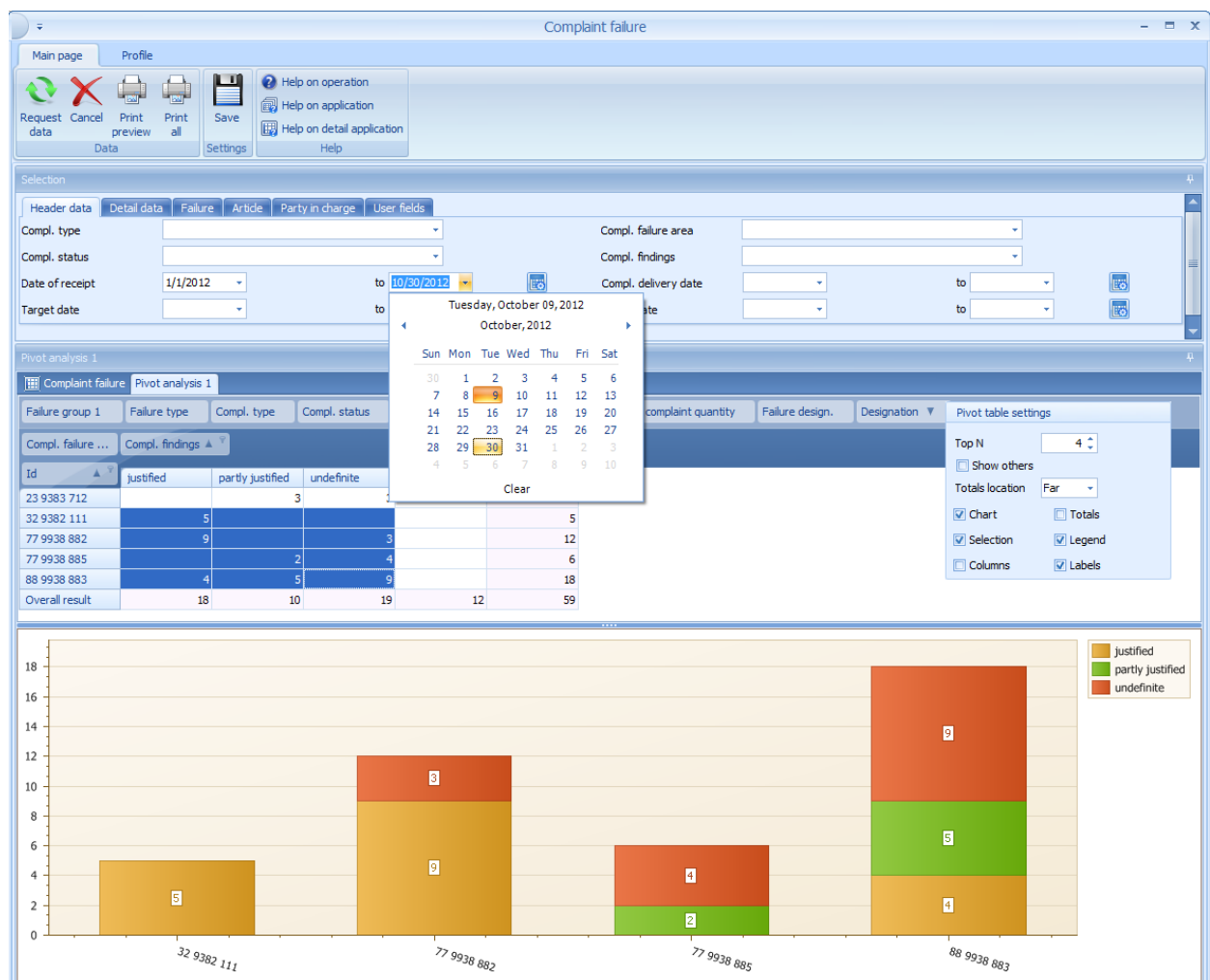
These functions are available.

- Graphic evaluation of recorded failure types, locations and causes using extensive filter criteria and presentation options
- Central and comprehensive measure tracking including editing options (status, efficiency, degree of fulfillment, actual dates, comments on performed activities etc.) and extensive filters (e.g. by status and responsible person).

2 Failure Analysis of Complaints

Overview

Menu	Quality management → QM evaluation → Complaint failure
Transaction code	faepcm
Function authorization	feapcm



Utilization

The failure mode analysis allows for failures of the following types recorded in complaint management to be evaluated:

- Failure type (FT)
- Failure location (FL)
- Failure cause (FC) and

- Causer (VU).

In this context evaluations/reports are based on pivot functions. They allow for the distribution of failure types (frequency) to be presented for each article/item or by the complaining party referring to a period of time that has been filtered beforehand. These analyses help determine the core areas that might require action to be taken.

Integration

The failure analysis of complaints only evaluates failures from complaint management.

Prerequisite

There are no special requirements to be met. The only prerequisite is that failures have to be recorded, which, in turn, need to be defined within the master data of quality management before.

Selection criteria

Selection criteria are self-explanatory and not described separately.

Toolbar

There are no other special function buttons in addition to the standard functions/features.

Detail applications "Graphic failure analysis of complaints"

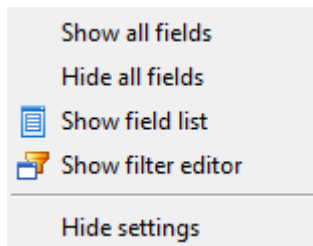
Data is displayed in a pivot table in combination with bar charts. Different application functions are provided for the presentation. The failures/defects that have been restricted beforehand by entering selection criteria represent the data basis.

The general pivot functions are not described in more detail in this document. The paragraphs that follow only describe the elementary functions of this evaluation/report.

Pivot evaluations/reports provide the following benefits.

- Large amounts of data may quickly be summarized and presented.
- Rows and columns can be exchanged to have the source data summarized differently.
- Simple filters by "drag and drop" with additional detail filters.
- Due to this interactive way of representation, data can be summarized and analyzed in different formats and using different calculation methods.

The below context menu can be opened by clicking the right mouse button.

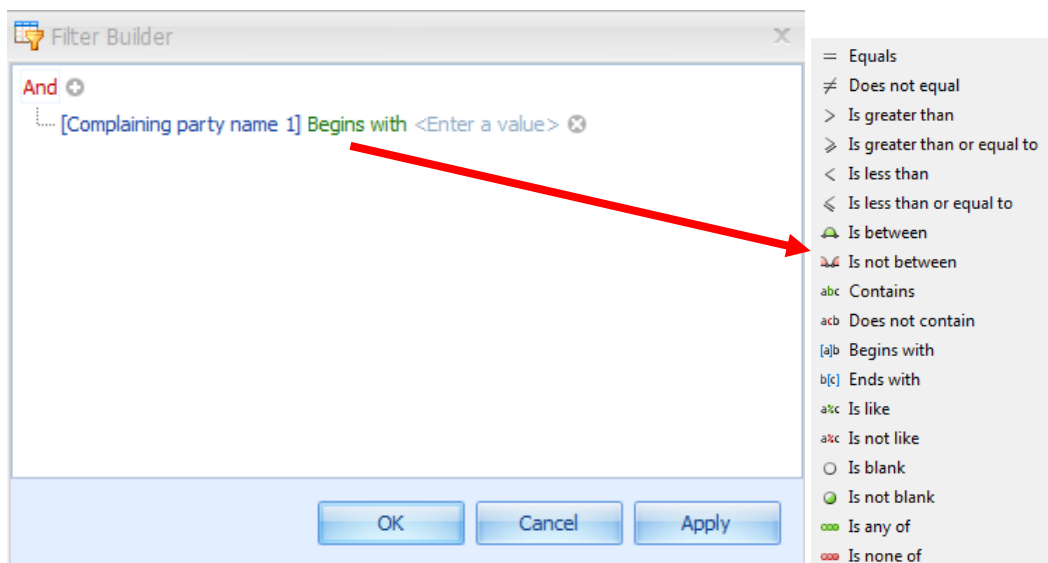


The function "show field list" allows for the fields that are to be used in the pivot analysis to be selected. The below figure shows a possible list of fields.



The requested fields may be put into the evaluation area by drag & drop.

In addition to the selection criteria, the "show filter editor" function enables further flexible restrictions of the data basis.



The below dialog is opened to show the settings made.

Pivot table settings

Top N

☐ Show others

Totals location

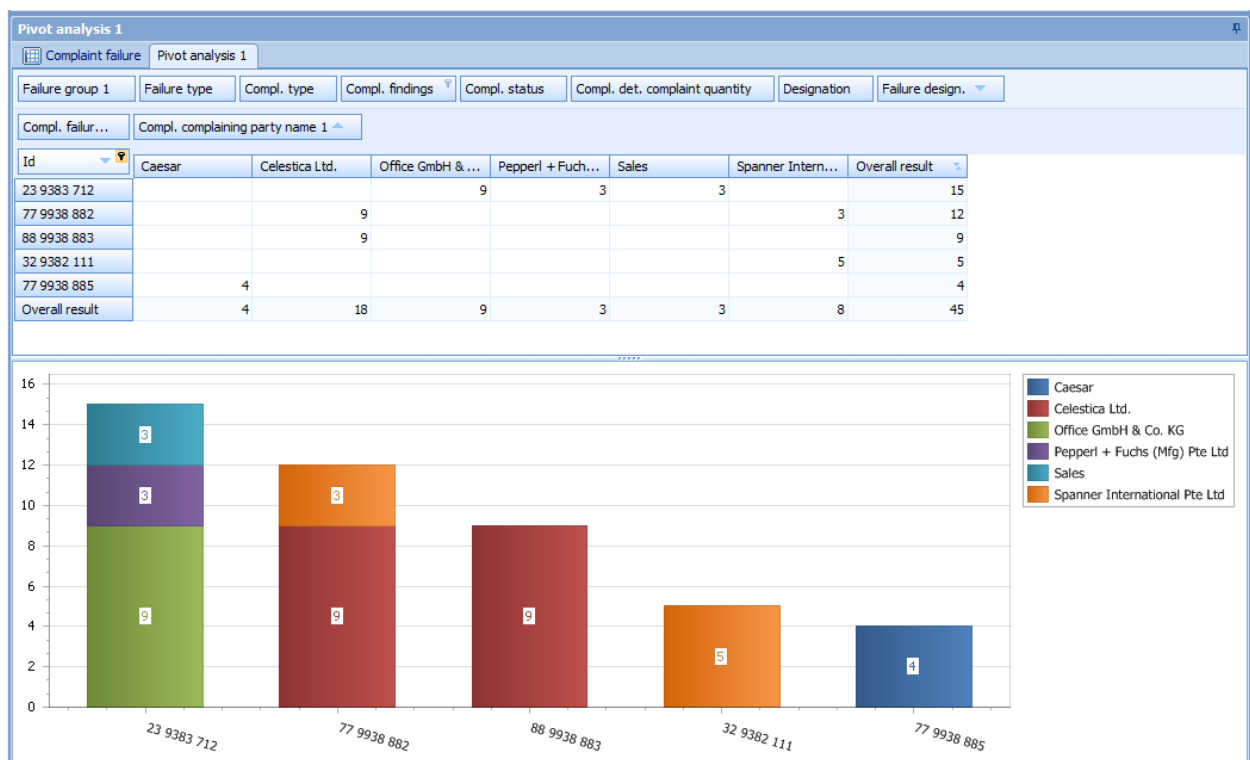
☒ Chart ☐ Totals

☒ Selection ☒ Legend

☐ Columns ☐ Labels

If the "selection" option is checked entire areas may be selected in the table view. In this case, the graphic representation is based on the selected rows. If the "label" option is checked it is possible to show the total number of each bar.

The below figure explains these functions.



The row showing the total result may be displayed additionally in the bar chart if the "totals" option is checked. Provided that the "selection" function is checked and the corresponding cells of the "total result" row are selected, the total result is added to the corresponding column of the relevant bar.

It is switched between the graphic presentation of the corresponding number of columns or rows by checking/unchecking the "columns" option.

Detail applications "failure list of complaints"

The failure list shows the failures including referenced data that are filtered on the basis of the used selection criteria. Normally, the referenced data correspond to the field list for the pivot analysis.

3 Measure Tracking

Summary

Menu	Quality management --> QM evaluation --> Measure tracking Quality management → Complaint management → Measure tracking
Transaction code	tm
Function authorization	tm

The measure tracking function provides an overview of all generated measures. The "type" specifies from which area/application the measure derives. The application allows for measures to be edited at a central place and assigned statuses to be changed.

Utilization

Filtering by the "status" is the central element allowing to select, for example, all measures that are open, done, read or in process. If this is combined with filtering or grouping by the party responsible, it is possible to create a list showing all measures that are still open by person, department, etc.

Measures may be changed. In addition to changing statuses, it is also allowed to enter a comment about the activities made in the fields "measure text" and/or "comment". Furthermore, this application also includes the actual date indicating when a measure has been completed and the degree of effectiveness and fulfillment in %. It is even allowed to change the party responsible.

The "reference" group provides information on the origin of the measure. If it is, for example, a measure generated in complaint management, the type, area and the complaint or complaint detail will be shown here.

The relevant application can be opened by clicking the corresponding button in the toolbar. The application is determined by the content of the "context" field.

The "long-term data" tab is only available if the archiving license has been purchased. If this option is set, long-term data will also be used when measures are displayed and tracked.

Integration

The measure tracking function is connected with all applications where measures are documented.

Prerequisite

There are no special requirements to be met.

Selection criteria

Except for the "reference" tab, selection criteria are self-explanatory and not described separately. The filter fields "key 1" up to "key 5" of the "reference" tab allow for measures to be restricted specifically to the origin of the measure. In case of a complaint measure, the field "key 1" includes, for example, the complaint number generated automatically in HYDRA. The key field 2 then includes the number of the complaint detail.

Field descriptions

The display dialog shows different tabs. However, if newly created/edited, there is only one tab (the "measure" tab) that includes all pieces of information.

Measures tab

Measure

Measure number and/or selection or direct entry of a measure number. The relevant master data catalog can be opened for selection purposes.

Measure designation

The designation of the assigned measure number is shown. If the measure number is input directly, the designation will only be shown upon saving.

Text

Free text field to enter a complementary measure text.

Comment

Free text field to enter a complementary comment for the measure.

Status

Available measure types can be displayed or selected. The types "in process", "read", "done" and "open" are provided by default.

Measure type

Available measure types can be displayed or selected. The following types are available by default "short-term", "medium-term", "long-term" and "no assignment".

Fulfillment [%]

Fulfillment in % can be displayed or entered.

Effectiveness [%]

Effectiveness in % can be displayed or entered.

External

This field can be used to control the printout of forms. However, it does not have a special function.

Context

Assigned object, e.g. inspection requirement, inspection step, complaint management, complaint detail, energy management

Type

The referenced application type can be displayed or selected, e.g. in-production inspection, goods receipt/goods issue inspection, initial sample inspection, test equipment inspection, complaint management, energy management

Area

The area of the type can be displayed or selected, e.g. production for in-production inspection

Key 1 to 5 (general)

The key fields are assigned depending on the context. These fields are empty for measures pertaining to energy management.

Key 1

Shows the internal and unique complaint number for measures from complaint management. The inspection requirement number is entered here for measures of inspection requirements or inspection steps.

Key 2

Shows the complaint detail number for measures from complaint management including assignment to the complaint detail. The inspection step number is entered here for measures of inspection requirements or inspection steps.

Key 3

Shows the internal, unique ID number of all failure analysis entries of a complaint for measures from complaint management that are also collected in failure analysis. The OP sequence number of the characteristic is entered here for measures of inspection requirements or inspection steps, provided that data in relation to the characteristic has been collected.

Key 4

This field is empty for measures from complaint management. The sample number is entered here for measures of inspection requirements or inspection steps, provided that data in relation to the sample number has been collected.

Key 5

This field is empty for measures from complaint management. The measured value number is entered here for measures of inspection requirements or inspection steps, provided that data in relation to the sample number has been collected.

Party in charge tab**Party in charge type**

Different types may be selected when inputting data. The type "external person" is used, as normally people are responsible for dealing with measures.

Party in charge

The party in charge is shown or it may be chosen from the list of responsible parties. Which entry is transferred to the list of responsible parties is defined within master data. The selected entry is used as the responsible party.

Party in charge name 1, name 2, name 3

The contents of the fields name 1, name 2 and name 3 of the responsible party are shown. The customer name and the content of the address fields 1 and 2 are displayed for customers. The last name, first name and initials are displayed for external persons.

Dates tab**Target date/time**

A date and optionally a time by which the measure has to be finished may be displayed or entered. It will not be monitored automatically, whether or not this time limit is kept. The content of this field is fundamental to the "measures tracking" application, as it may be determined manually for all measures (global) which of them have exceeded the target date, for example.

Actual date/time

A date and optionally the time by which the measure has been finished may be displayed or entered. However, this field does not have a special function. In the "measure tracking" application, this field can be used to manually determine measures that have been completed with delay.

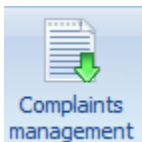
"Reference" tab

Area

Shows the referenced area within the application type, e.g. production

Toolbar

Apart from standard functions, the referenced measure, referenced inspection requirement or the consumption analysis can be called up for the measures from complaint management.



Calling up of referenced complaints

Function authorization: cm.*

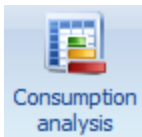
Calls up the complaint referenced for the selected measure.



Calling up of referenced inspection requirements

Function authorization: irp.*

Calls up the inspection requirement referenced for the selected measure.



Calling up of the consumption analysis

Function authorization: cona.*

Calls up the "consumption analysis" application

Editing functions

New measures can be created, provided that relevant skills are available. Consequently, users must have profound knowledge, as they are not guided through the process of data collection/modification and validation checking is not performed.

Relevant data need to be entered manually in the key fields 1 to 5 as well as in the context, type and area fields to enter new measures. The field descriptions provide further details about these field functions.

Please note:

New measures for complaints should be created within Complaint Management, as the creation within the Measure Tracking function requires the internal HYDRA complaint number to be entered.